

BITS & PIECES

* This month is the second month the all subscribers are on a new mailing system created by DBASE II. This system allows me to store ALL subscriber info on one disk, a much smoother system than previously used. Please check your labels to be certain all information is correct. Over 3000 names along with associated data had to be re-entered. Although the intermediate and final lists were checked and double checked, there's still chance of error.

This new system prints the word RENEW on your label when 1 more issue remains, and LAST with your last issue, to give you more of a renewal warning than before. To save time and money I do not send out renewal notices until you have missed an issue. If you do not renew within 2 months of your last issue, you are automatically dropped from the H-SCOOP list and placed on the nobody special list.

Some subscribers may have noticed from the label that you have extra subscriptions, and this was not a mistake. When re-entering the list, I gave some individuals lifetime memberships and some received a free extended sub. This was determined by past contributions to H-SCOOP, especially the software offers. Other factors were also involved in this decision. Some individuals have even received royalty checks for past and present contributions, again, especially software. This will be evaluated periodically and checks will be issued occasionally and without notice.

* A few notes on sales policy of QUIKDATA. TM-100-4's are lowest priced at \$365. These are single and quantity prices up to 6, and CASH prices. I am not able to offer any decent quantity prices since the prices are too low already. I am also discontinuing my practice on sending items out by "trust" and promise to get a check in the mail to me. My volume and types of items handled are getting too high and mass confusion is abounding around here. I have also been burned too many times from too many firms, mostly schools and colleges on Purchase Orders. No more purchase orders. Terms on ALL items: Check for those I know, Certified or Cashiers check for those I do not know, no COD's, no Purchase Orders, and no advance shipments. VISA and MC will continue to be accepted at a 3% increase. The charge companies make me pay an extra 3% for handling any card orders. My prices are based on cash sales and I cannot absorb the extra 3%, which will be passed along to you.

* DBASE II has been selling very well. There is more and more items filtering in every day about the capabilities and praises of DBASE II. Anyone interested in forming a DBASE II user's group, DUG, let me know. We may be able to exchange ideas, information, support, and programs. It looks like the best and hottest software item to hit the micro scene. For those interested, I should mention that 56K is needed for Heath machines. My price is still \$495.

* Charles Bennett called to report on the CP/M for the Percom DD H89 controller which he recently received. He says you can operate it in double density mode for 148K (40TK SS) and 624K (80 TK DS), or Extended DD mode for 186K and 782K respectively. It includes a new monitor ROM with many features, and many more to be added in the future. The BIOS can be configured for 48 or 64K, and you end up with 2 BIOS's on the disk, perhaps one for single density and one for DD. You can only boot

on SY0: 40TK drives, either SD with the Heath controller or DD with the Percom controller. Charles tells me that under CP/M (not HDOS) it will read a Heath DD 40TK CP/M disk made from the H37 controller.

Interesting, a day after Charles called, Paul Kelley called about the board. He said their CP/M BIOS DOES NOT support their parallel port on their disk controller card! How absurd! Paul also said he called Percom since he wanted to know if he could use their DD controller stand alone (without the Heath controller) and was told he could. He purchased it only to find he had to also purchase the Heath controller before he could use it, as Percom had been saying all along! It's like I said before, Percom does not appear to have their act, or at least their communications together. I would be very cautious about dealing with a company like that, no matter how large they may be. The bigger they are, the harder they fall, and the more they take along with them!

Paul called back later to tell me he had found a way to use his system with only the Percom controller (no help from Percom or anyone else). He says you only need the Heath controller to make up your disks, if you can find someone to do it for you, you do not need it. If the H17 ROM is placed in U520 of the CPU board, and the 4 pullup pins on Percom board are jumped, you can boot and use both hard sector and soft sector disks. Even Percom did not appear to know about this! If anyone has questions on this, contact Paul at 236 Varick Rd/ Waban, MA 02168.

* Last month I mentioned the CP/M selectric driver scheme and credited the work to Fred Freeland (pg 2). I want to mention that Claude Abbott was the software expert behind this BIOS mod for the driver, Fred is more involved with the hardware end. In fact, Claude has come up with lot of neat things under CP/M, including the type ahead buffer in version 2.0. Perhaps Claude will send us some unique CP/M tricks to publish in future issues.

* Note on a new address and phone number. Bruce Bennett of Mountain Software Co has changed their address as follows: 2620 Schuyler Rd/ Marietta, NY 13110/ (315) 673-2439. Micronet # 71435,200.

* Last month the phone number from Extended Technology Systems was given and published incorrectly. To get in contact with the company heads, contact George Deffendall at (215) 376-5043, David Lynch Jr at (717) 397-7406, or Robert Todd Jr at (215) 752-4604. They presently have available the Extended SYSCMD.SYS, Extended CCP for CP/M, and the H89 16K RAM add-on kit, as published last month.

* Comments from Bruce Bennett on the MBASIC Compiler. "Several month ago, you were discussing the MBASIC compiler which was about to be marketed by Heath. This version is to be compatible with HDOS. I use the version which is compatible with CP/M. It is a real disk memory HOG! A compiled program requires about 15K more disk space than the source code. Thus it seems to perform worse than Microsoft's FORTRAN compiler on this score. The program BASCOM requires a full CP/M formatted disk leaving no room for the operating system. This makes it inconvenient as the system will not reboot after BASCOM is run. . ." That's the point I made long ago. Most of these compilers have too many disadvantages. I'll stick to Pascal.

* Bob Todd called to tell me that Floppy Disk Services

came up with a way to get a **faster step time** on the Siemens drives. If you recall the old Wangco's would step at 6 or 8, and when Siemens took them over, they were limited to about 12 to 14 ms, with a guarantee of 30 ms. I have been told that if you change R48 on the drive board to 20K (is presently about 33K) the drives may be able to step much faster, like 8ms or so. The resistor value was changed in the first place to not drive the stepper motor so hard, and the stepper was also changed. Try it and send me some feedback. It may be easier just to jump about a 40K resistor in parallel with R48.

* For those who upgraded the H19 to an H88 don't throw out the transformer and PC board. You can construct some nice power supplies with this. I used mine to make a 5V @ 2A and 24V @ 1A supply for an 8" Tandon thinline. You could also power at least one 5" floppy from it and have power to spare.

* All vendor direct mail advertising blurps have gone out. If you are a vendor and did not receive one, and are interested, contact me.

* We are developing a **dual horizontal** (side by side, about same size as H17 dual cabinet) **5" floppy cabinet** and power supply. Will be a good unit, and not require a fan. Totally closed top means you can stack stuff on top of it. Got fed up with waiting and no one coming out with any decent dual cabinet. Expect a price of around \$150 or so. Anyone interested, contact me to get your name placed on the waiting list. I hope to be delivering these within a month or so. It is the same cabinet I will use for the 10 meg hard disk for around \$2000 I mentioned last month. Dealers interested in a good cabinet, contact me.

I am again in supply of single horizontal disk drive (5") cabinets with power supply. They are excellent quality and will sell for \$90. This includes cabinet and power supply with 4 pin connector plug. I am also restocked with 34 pin disk drive cables.

* A new place, **Independent Technical Service/** 18925 Gualt St/ Reseda, CA 91335/ (213) 705-3604, was gleaned out of Computer Shopper for disk drive repair service. They have a 48 hour turn around policy (free loaner for late delivery), and charge \$50 for any 5", or \$75 for any 8" drive. This price does not include head or stepper motor if needed. 30 day warranty for service.

* For those using the 5.11 (newest release) of **Spellbinder**, Lloyd Johnson offers an enhancement package. Lloyd is an engineer interested in Human Engineering and has changed some of Spellbinder's HELP messages and key layouts to a form he considers easier to learn and use. For example, in the EDIT mode labels now appear above the special function keys of the H89 indicating their purposes. The names above the labels are more in agreement with terminology in the Spellbinder manual than in the messages provided by Lexisoft. His changes also permitted extra room for listing most often needed Dot commands. Some changes were also made for the COMMAND mode. There is more, but if you want all the details, contact Lloyd.

He made the changes by modifying the IOS file, assembling it and inserting the resulting HEX file in H89SB.COM to produce a new unconfigured SB file, all according to instructions in the section of the manual on SYSTEM INTERFACE. Since many are not familiar with the IOS or assembly procedures for CP/M, Lloyd is making the changes available to all interested. In order to comply fully with the terms of license agreements with Lexisoft, Spellbinder owners must send Lloyd a disk with copies of two files from their copy of Spellbinder 5.11 version, plus \$15. The files are HSB.COM and HELP.HEP made from their distribution disk. Lloyd will make the necessary changes and assemble it for you, send

along some excellent sheets of the HELP.HEP commands, instructions on how to do it, an unconfigured but modified version of Spellbinder, and all other related goodies. Contact Lloyd at 700 Highview Rd/ E. Peoria, IL 61611/ (309) 699-1476.

* A very upset Bob Iacullo called me several times and wrote about some problems he is having getting his H89 from the NTS computer course. It appears NTS has been telling Bob (and many others) since October that some of the parts have been on back order from Heath. Bob then started calling Heath who kept back ordering the kits "for another two weeks", and this is still continuing. The rest in Bob's words. "Now I know that the parts are available because they are shipping H/Z89's out to the dealers all the time. The only difference between these computers is that the NTS 89 has less RAM, and no serial board."

"But since my H89 is already paid for and at bulk price not list price, I believe that their attitude is that I can just wait until they feel like getting around to me. They apparently don't care about supporting customers once they get their money. This is no way to run a business and I can assure you that I will not buy from Heath again."

It seems that the attitude of top management is the hobbyist does not require (or deserve) the support the business system users require. **NOT TRUE!** I have been receiving many complaints here about common parts such as drive cables, RS 232 connectors, etc not being available from local Heath stores or from the factory. Yet as Bob says, I know they are shipping out large numbers of systems daily, so the parts have to be there! I have a dead H89 sitting here, thank God it's my spare. I built the H19 to H88 conversion kit and had a bad ROM. I installed the Magnolia DD board, and it had a bad ROM. March 26, a call was placed to Heath and Magnolia. Three days latter I received the Magnolia ROM, but have yet to receive the Heath ROM, it's on "Back Order"!! So here I have \$6,000 tied up in a system with 8" drives and all, ready to go, but down for a \$10 "back ordered" ROM, which they do have for the new units they are selling.

I take this space and time to make this point because I feel hobbyists DO deserve support. My policy here on drives and what I sell, is service the past customers problems BEFORE new sales are made. If I have a customer with a bad drive, a new one comes off the shelf and goes out to him while the NEW customer waits. I also keep my stock up so the New customer usually doesn't wait either. That's only fair. Support who you have already out there. Keep them happy. That's the best free advertising anyone can get. I don't like to be hard on Heath, but this back order situation is really getting out of hand. Just because a kit was built does not mean the customer is not doing business work with it. I hope Heath takes note of this and does something about it before it gets much worse. I felt I was forced to buy a second system because of this support and back order problem, and one should not have to feel this way.

But don't only call or write to me, I'm not Heath. If you feel this same way, get off your duffs and write Bill Johnson at Heath and let him know how you feel. He can do something constructive about it, I can only rattle the cages a bit.

* Jim Benesch sends in this short bit for users of the new Trionyx Z80 CPU. "The 4 MHz LED on the Trionyx Z-H8 can be relocated to the H8 front panel by soldering two wire wrap pins in D1 (the spacing and hole size is exactly right). Using a connector, (like the one which isn't needed if you are using the bus for side select), and 6 inches of wire, install the LED (observe polarity) in the unused location (already drilled and plated) on the top left hand side of the H8 front panel."

* Ray Livingston of Livingston Logic Labs has released an updated version of the **BIOS-80** for CP/M 2.2.03. The real difference between this and the earlier release is this only contains one BIOS. The earlier one contained a 2 MHz and 4 MHz Bios. You had to pick the one you wanted to use and rename it BIOS.SYS. The new version will automatically adjust to CPU speed, no additional ROM's required. Thus if you operate your system on both 2 and 4 MHz speeds, this one will take care of it all. Same price, \$35, or update from any previous version for \$10. I will handle this, Ray Livingston, and Ray Massa of Keyboard Studio.

While on this subject, I want to stress there is absolutely no problems with this CP/M 2.2.03 **BIOS-80** from Livingston. Calls have been coming in because of the **SPECIAL NOTICE** on page 16 of **REMark** issue 28 (May '82). Lot of people are reading in to this that there is a problem with it thus HUG does not care to support it. Not true! There were other reasons why HUG is not handling it, the quality of the product had absolutely **nothing** to do with it. In a nutshell, both parties could not come to agreeable terms on the contract, so Livingston pulled the BIOS-80 from HUG.

* Good news on compatibility for the Magnolia Double Density users. There will shortly be released additional modules for their present CP/M which will allow your software to be compatible with the H37 and H47 Heath formats. We now have this under HDOS, the CP/M compatibility will be very welcomed. On a slightly different subject, Magnolia's new release of CP/M will be awhile yet.

* Most of the new Tandon drives shipped now have a door switch added which starts the motor on door closing to help center the disk, thus eliminating the crunched center problem for careless users. Mike Miller of Walt Disney World called to tell me that if you are using a **ONECOPY** type program (Available from Viking Software as CP-80) on the Tandon drives, you must be careful. This onecopy program requires you to do lot of disk swapping, which requires many door closures, which activates the motor. If you are too fast and don't wait for the motor speed to stabilize, you will get errors. The motor starts spinning quite fast and requires about a second to slow down after the door is closed before the drive should be accessed.

* Larry Reeve of Polybytes tells me the HA-8-3 support package for the Lucidata Pascal V3.8 will be ready to ship by the end of May. It is available for both HDOS and CP/M. There are 53 procedures and functions available to provide a high level of control of the A/D, graphics, and sound generation, with all source provided. Sophisticated procedures are provided for point plotting, high speed line drawing, sprite manipulation, and an arbitrary shape painting procedure that must be seen to be appreciated. Price is \$35, or \$15 if ordered with the 3.8 Pascal system. Contact Polybytes for more info.

NEW STUFF

SOFTWARE RELATED

* Pat McNally wants to add to NEW info as published in the last 2 issues on **WORDSTAR** that his enhancement package allows the WordStar system to be used **without any control key strokes whatsoever**. The normal control keys still work for those who want to use them.

* Another from **EVERYWARE**/ POB 60802/ Sunnyvale, CA 94088 (415) 321-2708. **SPACE ODYSSEY I** by Joe Gargiulo is a 3-D graphics game. Sounds excellent! You are in the cockpit of a ship in the year 2033, exploring

the **ORION** cluster. A variety of objects come at you in 3-D space, as you look out the front window of the Atlantis. Use flight computer to find your heading, receive navigation messages and get distance readings. Many more features and of course, many obstacles. H19 or H89 screen needed and 48K CP/M or HDOS. Available at most HEC's on 5" hard sector for \$21.50, or from **Evryware**. Comes on Dual format CP/M & HDOS disk. See **REPORTS** for more on **Evryware's** quality software.

* **ZenCalc** is a spreadsheet calculator available from **Software Toolworks**/ 14478 Glorietta Dr/ Sherman Oaks, CA 91423/ (213) 986-4885. It is a tool for financial and numerical model building, forecasting and problem solving showing you the **WHAT-IFs** in a situation. The program features use of labels, formulas, variable column width, and can print sheets to a printer or disk file. As with all of Walt's products, the special function keys make it a breeze to use. Manual includes a step by step hands on tutorial. Examples are included, and "templates" are supplied to automatically compute the full IRS form 1040 and itemized deduction schedule A. Available on 5" HDOS or CP/M disks for \$99.95 from most HEC's or from **Software Toolworks** (add \$2 postage and handling).

* **EDBS**, **Excalibur Data Base System** under CP/M is available from **Software Wizardry, Inc.**/ 122 Yankee Drive/ St. Charles, MO 63301/ (314) 946-1968. **EDBS** is menu driven and takes advantage of the H19 style display to lead the operator 'by the hand' through the creation and maintenance of a database. Because ASCII, numeric and relational field types are supported, **EDBS** can manage file space in an intelligent manner to store the largest amount of info in the smallest possible space. Information in the database can be searched and sorted on ANY field, sorts being handled by an **.ABS** routine for speed. Up to 32000 records can be handled depending on disk space and record size.

The included report modules allow the user to output stored info in any number of different ways. The data can be used to create mailing lists and labels, form letters, charts, graphs, files, and even new database source files.

Requires 64K RAM and CP/M 2.2, along with **MBASIC** and at least one drive. Price is \$49.95, **VISA** and **MC** accepted. HDOS version to be released soon. Watch **H-SCOOP** for details. FYI **EDBS** was originally sold in Canada for over \$120.

* **COMPUTER CHEF**, an electronic cookbook and recipe file is available for \$29.95 from **Software Toolworks**/ 14478 Glorietta Dr/ Sherman Oaks, CA 91423/ (213) 986-4885. A cookbook on a disk, **COMPUTER CHEF** features auto recipe search and ingredient scaling. It comes with over 70 home-tested recipes. Personal recipes can be added using most editors.

Recipes can be searched by ingredient, title or keyword, to select, for example, all quick chicken recipes without salt. Available on 5" and standard 8" CP/M disk.

* Unit III of Gene Gingerich's **Handbook for Heath Users of Softech's UCSD Pascal**, version II.0 is available and includes how to use the assembler with examples of how to construct assembly procedures and functions to interface with Pascal programs under Softech's system. Also includes how to handle parameters passed by name or value, and how to link them using the **Linker** and more. Cost is \$4.00 plus \$.50 for shipping in US, or \$2.00 shipping foreign. Order or obtain more info from **GINGER BREAD SOFTWARE**/ POB 1627/ Santa Barbara, CA 93116/ (805) 685-6556.

* **FOOTNOTE FORMATTER** is an MBASIC program that takes your text file and generates a perfectly formatted thesis style output file with footnotes. Conforms completely to Turabian's "Manual for Writers." Essential for students! Works best with Software Toolworks **TEXT**, but any editor can be used. **NOTE CARD GENERATOR** speeds the production of note-taking. Program produces indexed and titled 4 X 6 notecards with full bibliographical data. Both for \$25 from RAY ALBREKTSON, 619 W Dover Dr/ San Bernardino, CA 92407.

* Early purchasers of the **ALBREKTSON CLOCK & SOUND board** for the H89 can get the latest **sound & time software updates** (such as touch-tone dialer program, patch for Micro-Piano, etc.) by returning their distribution disk and \$1 to cover costs to Ray Albrektson. Ray also mentions that he is still selling the assembled and tested version of the board for \$90; bare boards with instructions and software for \$27.

* **T & E ASSOCIATES, INC/** POB 362/ Millersville, MD 21108/ (301) 987-4748 has a variety of software available. It may be important to note that T & E represents computer professionals with expertise in large mainframe computer networks. All of their software is written in tightly coded assembly language and is human engineered and configuration controlled. They are probably the leaders in printer software drivers, all containing many outstanding features.

Briefly I'll list the software they have available, all under HDOS. Write to them for more information if desired. **TRS** for \$40 is a TRS-80 to HDOS file converter. A soft sector disk from a Model-1 TRS-80 can be read on a Heath H17/ H77 and transferred onto Heath hard sector format. It will also handle TRSDOS-80 and NEWDOS disks. **DKSDUP** for \$15 is a stand alone utility that duplicates and verifies disks (40 track). **FORMAT.ABS** for \$15 (includes source code) can be used with HDOS in place of INIT to save disk space since only 12 sectors are required. **TAPE I/O** allows you to store files on cassette (H8 or H89 cassette interface required) using HDOS. You can store any files including those that are too large to fit on a single disk. The files can then be read to a printer at any time. Important disk files can be stored on tape as a backup. There are no restrictions on the type of disk file that may be written to or read from tape. **TAPE I/O** is not compatible with the files created by the Heath Cassette Operating System without prior conversion. **INTERACTIVE JOB UTILITY** for \$35 is a sophisticated **SUBMIT** package that gives you 'batch' processing capabilities. Unique features are ability to start, suspend or abort the processing as well as log all console activity. There is no restriction on size of a command file or number of commands.

The rest of their present software deals with device drivers. **MULTI-PRINT** for \$40 is a package to allow you to spool data or files to four separate devices, all running at the same time. A unique feature of Multi-Print is the ability to Start, Stop, Abort, etc. any and all spooled channels, even while running an application program. Many of their drivers have **ECOSSET** (c) to display the current SET option values. Each spooling device can 'queue' up to 15 files for output. **BIDIRECTIONAL DEVICE DRIVERS** for Diablo and NEC printers allow the printer to print bidirectionally for smooth operation and maximum speed. Also features **ECOSSET**. Escape codes can be embedded in the text file being printed to change option values (normally changed using the SET.ABS HDOS program). Other features include Custom initialization features, left and right margin settings, immediate printer suspension without data loss, typewriter emulation, Multi-Print spooler compatible. Requires XON/ XOFF (DC1/ DC3) protocol. Cost is \$25, specify Diablo or NEC version.

MAXI-PRINT DRIVERS for \$20 each are available for the following printers: LPH14ES (H14), LPH24ES (TI-810), LPH25ES (H25), LPH34ES (DIABLO/ NEC), LPH44ES (DIABLO/ NEC), PT (PAPER TIGER), MX (EPSON), and OK (OKIDATA). These drivers provide maximum speed while using minimum RAM space. All drivers feature **ECOSSET**. Brief summary of other Maxi-Print features include pagination, custom initialization features, left and right margin settings, immediate printer suspension without data loss, typewriter emulation, and are Multi-Print spooler compatible.

Software supplied on 5.25" SS SD 10 hole hard sector 48 TPI disks.

HARDWARE RELATED

* Update on products available from **MI-8/ 822 E** County Rd 30/ Ft Collins, CO 80525/ (303) 669-4116. **MODEM** kit is a direct connect modem which operates at full duplex at 300 baud. The modem with complete software package and assembly manual sells for \$99.50, while the kit without software sells for \$64.95. Software includes many user options including auto log-on for fast access to Micronet and the **HUG BB. COMMAND-8** is an H8/ H89 voice recognition and control system. Requires Z80 processor and parallel input port, with 48K system. Easy to use, the recognition accuracy can be compared to voice recognition systems costing far more. With **COMMAND-8** you can add voice control to your BSR controller and simply **TELL** the computer to turn a light or appliances on and off. You can also speak to your computer and have it verbally respond with the **V-8** voice synthesizer. Comes completely assembled and tested ready for installation, and includes documentation and software. Price is \$99.50 until June, when the price becomes \$124.95. Guaranteed.

The **BSR HOME CONTROL INTERFACE** for the H8 and H89 computers allows your Heath computer to control up to 256 lights, appliances, motors, or any electrical switches or devices. Ultra-sonically coupled and completely safe, requiring no special wiring. Just plug it in. Comes in kit form for \$69.95 or assembled and tested for \$89.95. **V8 VOICE SYNTHESIZER** model II sells for \$249.95 kit, or \$299.95 assembled and tested (includes assembly and operation manual and demo and diagnostic programs on 5" HDOS disk). This is the only voice synthesizer for the Heath systems (requires parallel port) which has automatic inflection and it can be programmed to allow it to speak in a foreign language. It has an unlimited vocabulary, which is user defined. You are not restricted to the flat mono-tone voice output like other synthesizers. **MI-8** also handles the **PC-12** parallel port board for the H8 and the **PC-89** parallel port for the H89, both by Environmental Control Systems. They have some of the lowest prices I have seen on **MX-80** printers, \$474.95. **Heatsinks** for the H8 boards are \$2.49 each. **DIGITAL TALKER** kit is \$119.95 and assembled version goes for \$149.95. I will be evaluating the Voice Command system and printing it in the next few months.

* Update from **Environmental Control Systems/ 9319** Willowview Ln/ Houston, TX 77080/ (713) 464-1717. The **PC-12A** is an assembled and tested 12 port parallel I/O interface card complete with perpetual clock and calendar for the Heath H8. Complete with extensive doc and software for HDOS (CP/M on the way). This sells for \$249. It's cousin, the **PC-89** is the same thing for the H89 only in a 9 port version (mounts in left side of the 89). Both come with self charging **NICAD** battery for clock. **RX-10** a universal wireless BSR X-10 remote controller is available at an introductory price of \$124.95. It provides automated and remote control of lights, appliances and all electrical loads. Provides all

necessary communication between ANY computer and the AC line to command the BSR system X-10 remote switches sold by Sears, Radio Shack and others. The newest release is the **RL-10**, which is a two channel stereo high performance voltage controlled attenuator. Available as a plug-in enhancement to the BSR X-10 remote control system to provide remote control of the volume of any stereo sound system (easy to install with amplifiers using 'preamp out- amp in' or Tape monitor In/Out facilities). Priced at \$49.95. They also carry a complete selection of cable assemblies for connections with their PC12A and PC89 to RX-10 controller.

* Update from **TRIONYX ELECTRONICS INC/** POB 5131, Santa Ana, CA 92704. Their H8 Z80 CPU has been well received and is available fully assembled for \$300. The complete parts kit is \$250, or the bare board for \$125. All of these include complete doc, HDOS and CP/M speed utilities on disk, and FW-Z80 front panel monitor ROM. Operates at 2 or 4 Mhz via system command or software. Those operating with the Trionyx 64K memory can get their memory board modified by the factory for \$50. This mod guarantees no wait states at 4Mhz, and will also work at 2Mhz, but no longer with the 8080 CPU's.

They are also working on an H8 universal floppy disk controller to be shipped in the fall of '82. Orders are being taken at the prices of \$425 assembled or \$350 kit. More on this next month.

MISCELLANEOUS ITEMS

* I briefly mentioned a two book set last month **Your Fortune in the Microcomputer Business**. I did not have time to print the details. This book set is two volumes, paperback. Volume I is "Getting Started", about 300 pages long, and volume II is "Growth, Survival and Success", about 230 pages, both by Victor Wild. I have read these books and all I can say is **excellent!**. Being in the business of making a living off of micros, I have quite a few books in my library on this subject, but none were really complete, nor worth much. This book set was written by a man who knows what he is talking about, has been there and is still going, and has the rare ability to put his ideas, concepts and thoughts into words.

The first volume covers many ideas on making money with your micro; how to start out, different ideas, setting up a business, marketing, promotion and advertising, things to look out for, and much more. He covers topics such as starting a computer store, servicing, consulting, newsletters, modem services, software and hardware sales and more. What impresses me mostly is the list of references and people or places to get in touch with scattered through the book.

The second volume, you're there, now what? How to keep on going and continue in success. This covers more of the business end such as accounting, cash management, wages, maximizing profits, how to treat employees, how to survive the hard times and more. For anyone in the business full time, or wants to get there, or just would like to make some extra bucks to pay for his hobby investment, I very highly recommend this book set. To get a successful businessman's ideas and experiences in print is priceless.

Because of the great value, I will be handling the book sales for a time. They are \$15.95 each. or the set for \$24.95, which is quite a bit lower than the original intended price, and worth much more than that. This price will include shipping. If you order, please specify which volume you want, or the set. You can also get these directly from Victor Wild/ 1 South Fariview/ Goleta, CA 93017/ (805) 967-8444.

* From **CompuCover/** POB 324/ Mary Esther, FL 32569/

(904) 243-5793 are covers for just about any computer equipment you have including the H25 printer, H17, H19, H89, H8 or any other Heath printer. Write or call for price list.

* Colored vinyl **DISKSAVERS** are available for 5 and 8" diskettes from Rockroy Inc/ 7721 E Gray Rd, Suite 103/ Scottsdale, AZ 85260/ (800) 528-2361. These are colored vinyl sleeves for use with single diskettes, to replace the paper ones that come with diskettes. Available in red, blue, green and yellow. Prices start at \$.22 each at 500 quantities for 5", contact them for 8" prices.

OFFERS

I make all monthly software (H-Sxxx series) offers available on an as-is basis, in other words, no money back policy if not satisfied, mainly because it's too easy to copy discs. You take your chances. At my prices, you really can't go wrong. I don't think I've let anyone down in the past, and will continue to do my best not to.

The following is the current offer list available through **QUIKDATA, INC.** As policy, this listing will be printed quarterly.

HS-XXX OFFERS

HS-001 \$10 General Utilities & Toys
 HS-002 \$10 Word Processors & Editors
 HS-003 \$10 H19 Graphics & Printer pic I
 HS-004 \$10 H19 Graphics & Printer pic II
 HS-005 \$10 Labels, Random Files, Plant
 HS-006 \$10 Challenger
 HS-007 \$10 HDOS 2.0 Patches & doc files
 HS-008 \$10 Quikdata Selectric & misc drivers
 HS-009 \$10 Misc Basic Programs I
 HS-010 \$06 Multi Key Sort Routines
 HS-011 \$08 From the Uni of Houston
 HS-012 \$18 Disk drive exerciser routines
 HS-013 \$10 Multi Dup & Dump

The complete description of these appeared in April 1982 issue. If you did not receive this issue, write for a description.

PROJECT ARTICLES FOR PARALLEL PORT--WRITE FOR CATALOG

Included in this series is A/D/A converter, 32 channel on/off controller, 600 Baud LA-36 mod, Telephone computer interface and more.

PROFESSIONAL SOFTWARE

Spellbinder word processor and office management package. The best word processor around, makes use of H89/ H19 special function keys. This supports true proportional spacing, and is easy to use. It's what this newsletter is done with. Available in 5" hard sector only. Lists for \$495, my price is \$285.

DBASE II data base system by Ashton-Tate, \$495, compare at \$700. The easily programmable complete data base system for mailing lists, business systems, inventory, or whatever you want, that runs under CP/M. Available in 5" H/Z format, or 8" (56K required). Has the most features of any dbase program on the market. Handles files of up to 65000 records, sorts and indexes on ANY field and multiple fields, full programmability (IF THEN ELSE, DO WHILE etc.). Perhaps this program along with Spellbinder is all the businessman or serious hobbyist will ever need. 30 day trial period.

Multiple Device Spooler by Software Wizardry, Inc. for \$24.95. Excellent HDOS spooler to print files on up to 5

separate devices independently of each other without losing normal usage of your computer. Requires 2.0 HDOS and 24K RAM.

ZLYNK by Software Wizardry, Inc. for \$24.95. This is the do it all modem package utility at an affordable price! Will do anything other modem packages do, and then some. On screen clock, various protocols, spool to disk or printer, transmit and receive .ABS files, auto log-on and much more.

DDUMP by elektrokonsult in Norway for \$29.95. This and the following CP/M utilities by elektrokonsult are quite unique in that they work on ANY single user CP/M 2.X system with ANY diskette size and format, and many hard disks. All are menu driven and easy to use. DDUMP makes it possible to examine and patch any byte on any sector addressed by track and logical sector number - or addressed by allocation block number, both in HEX and ASCII format. May be used to study how CP/M allocates disk storage, to examine or repair a damaged disk and to recover lost data.

DTEST by elektrokonsult for \$29.95. This tests a disk for bad spots and locks them out from later use by CP/M. All errors are reported to printer and/or console.

DDUP by elektrokonsult for \$29.95 is a disk duplicator independent of disk controller, drive, disk size and format. It only requires that source and destination disks are of the same format and density. It will also replace bad sectors from the source with blank sectors on the destination, enabling you to automatically recover (what can be recovered) damaged files.

DKH17V2.DVD by Ultimeth Corp for \$40 (updates from SY version original disk (HUG disks accepted also) + \$20). Used with Tandon drives in a single density system to get 400K per 100-4 drive. Many exceptional features such as settable step times and motor control idle time. Great for 40 TK drives just for the features. For HDOS 2.0 only.

DKDDFLP.DVD for use with the Magnolia H89 DD board to allow HDOS to be used with it. Excellent HDOS driver for a CP/M product! \$40.

BIOS-80 for CP/M 2.2.03 for \$35 (updates from 2.2.02 CP/M version \$10 + original disk). Simple BIOS replacement which retains all BIOS 2.2.03 features plus allows you to use anything from 100K to 400K drives with it single density for 392K storage per TM-100-4 drive.

AND HARDWARE ITEMS

MAGNOLIA H89 DD SETUP. One of the best ways to go to get double density for your H89. Complete with CP/M cables and all needed. Supports 4-8" drives AND 4-5" drives in addition to your H17 controller! \$495.

TANDON TM-100-4 drives. \$365. Double side, double track 96 TPI disk drive.

TANDON TM-100-2 drives. \$290. Double side, single track drive.

TANDON TM-100-1 drives. \$225. Single side 40 track drive just like your Siemens.

TANDON 848-2 8" Thinline double sided disk drive for \$495.

SINGLE 5" CABINET (Horizontal) with power supply for 5" drives, \$90.

DUAL 5" CABINET with power supply for 5" drives, side by side horizontal, \$150 (tentative price).

All prices are cash, credit card add 3%, and includes shipping and handling for UPS delivery (the larger items). For mail orders add \$5 extra per item over \$200.

WI residents add 5% tax.

I also offer a **disk conversion** service. I can upload or download disk varieties from 40 track SS 5" to 8" DD DS Magnolia format, and standard SD SS CP/M IBM 3740 format and most anything in between.

I have separate catalogs, brochures, descriptions and price lists available for most of these products. If you desire further info on any item(s) let me know.

This months offer is **HS-012** for \$18. It is a comprehensive **DISK DRIVE EXERCISER** software set written in .ABS and runs under HDOS. This disk is definitely not for everybody. Anyone every trying to check out a disk according to manufacturer test procedures, or align a disk, knows usually a \$500 device is used called a disk drive exerciser. This is simply a device which connects to a disk drive and can position the head anywhere on the disk, and do various reads or writes. I reasoned that since most of us already have the computers, we may as well use them, and just get software to emulate the expensive disk drive exerciser. I contacted Terry Smedley, and HS-012 was born.

This is only for H17 variety drives, including the Tandons. Full doc files are present which fully explain the operation of the exerciser. The exerciser will allow the user to seek any unit, track, and side and do continuous reads or writes of constant bit patterns (1's, 0's, or arbitrary pattern). There are also facilities for checking the track zero detect, write protect, and index hole sensor hardware. A drive rotational speed check is also included. These tests should serve to locate the majority of drive difficulties.

If one has the maintenance manual, an alignment disk, and a good scope, he can also do alignments with this exerciser. The menu (H19/89 needed) includes the following: Select drive unit, Select track, Toggle side select, Seek track and read data, Seek track and write all 0's, Seek track and write all 1's, Seek track and write user specified byte, Seek track zero (home heads), Check write protect hardware, Check index hole detect hardware, Perform drive rotational speed test, select drive track step time, exit to HDOS.

As usual with the HS series, this is an exceptional piece of software by the same guy who did the HS-013 Dup/Dump programs. You'd expect to pay around a hundred bucks for this type of stuff. But there's more. Also included is an extended statistics program, XSTAT. Shows all the STAT conditions of most other ones on the market PLUS more. Dumps the SET options for the SY and TT device drivers. It also includes a very complete .DOC file (DSKERR.DOC) which explains in detail how HDOS handles disk errors and what the individual error types mean. Files on disk include XRSIZR.ABS, ASM, and DOC; XSTAT.ABS, ASM, and DOC; DSKERR.DOC and all needed .ACM files to re-assemble the programs, except for the standard HDOS ACM'S. In other words, all source in included! See you in three more months.

REPORT

EVRYWARE GAME REPORTS

I have been playing some of the H89/ H19 graphics games from Evryware Software (see NEW STUFF) and must say, there are some good action graphics available from them. I must admit they are definite leaders in H89/ H19 fast action graphics games. Thus far they had MISSILE COMMAND, INVASION, GALACTIC WARRIOR, and Y-WING FIGHTER, with SPACE ODYSSEY just out. They are all good graphic games, but I would have to recommend Y-WING FIGHTER and ODYSSEY as the

winners of the group. I have not yet made it to the end of any of their games. I and others have noticed an occasional bug in the graphics on Y-WING. It's nothing serious but it does come up once in a while. I talked to Joe and he assured me there was nothing in the software, but an escape sequence problem with some systems when many graphics escape characters are outputted in rapid succession.

Joe Gargiulo and Dave Murry are partners in the software company **EVRYWARE**. They are both programmers, and both are very knowledgeable when it comes to graphics and control of the H19 type screen. Their software is done in 'C', which compiles to an .ABS machine executable file for fast speed. Evryware has several unique trademarks which can be seen in all their action graphics games. There always seem to be surprises which you were not anticipating. The longer you play a game, the more things that happen; more obstacles, different creatures, etc. Also the longer you play, the more difficult the games become. There is also mystery about most of them. The instructions tell you enough to play, but there's usually more that you must discover for yourself. I don't want to give any of this away, so I'll leave it at that.

Their newest just released game, **SPACE ODYSSEY** is a real winner. This is definitely a challenging game. It requires much thinking, controlling, studying, and experience to go anywhere. Don't even expect a good score the first times through. You are piloting a ship in space and are looking out your front window into 3D space. At the same time you have to watch the heading indicator to keep the ship on heading to reach your data probes which were sent out to collect data a year before your mission. These data probes are then fetched by the tractor beam and taken aboard the ship. Another instrument to watch is the scanner, which shows a side view of the front of the ship with space coming toward it. Here you see how close objects are to hitting your ship. Use your laser to blast them before they hit you, maneuver your ship to escape them, or ignore them and they may go away. If you get hit, the shield energy indicator will decrease in its reading, when it's used up, all done! There's also a fuel gauge to watch, and a navigation and communications screen. The entire keypad is used for maneuvering; the function keys, and the QWER keys are used to control WARP speed, HYPERWARP to get you in and out of the Orion cluster, Gamma ray laser type gun, sights, tractor beam, status indicator and jettison. It's not as complex as it sounds though. Once you have played it for awhile, you know what you can ignore a bit and which ones to pay more attention to.

All in all, frustrating at first, then as you get to know it, better to excellent. Very exciting and challenging, and keeps you on your toes at all times. Would highly recommend it to ALL who do not have nerve problems, and who do have lot of time. **RATING A.**

On the first four games, D. C. Shoemaker writes in about those same games, "All four are good, and "Galactic Warrior" is unusually challenging, but the odds-on favorite in my home is "Y-Wing." Unfortunately, there seems to be a special quirk in it; of five H89's I tried it on, only two would support the graphics. No amount of fiddling with the terminal options seemed to make any difference. Works fine on three H19 terminals I ran it on. Very Puzzling." Thank you DC.

Back to me, **Y-wing** is very challenging and exciting. Different obstacles pop up at different stages of the game. I never got much past the volcano, but another I talked to was all excited about the helicopters, and later when he made it further, he was really excited about the spiders (I thought he was going to have a heart attack--I had to calm him down). In this game you are flying into

the earths orbit and must make your destination, over 500 miles. I have never made it past 200, but others have made it over 400. I thought Munchkins was great, but I think for excitement this is better since there is something different as you go farther and become more skilled. With a game like Munchkins, although it's great and the graphics are good, it's always the same (but you can change mazes and parameters to make it different). Y-WING has excellent graphics, and I would recommend this to ALL.

I also tried **SIX MICRO-STORIES** (\$17.50). This like all their other software runs under HDOS and MBASIC. You start off with a story and go through all six, answering questions. Your response to the presented question determines the next part of the story, and ultimately the end. Since this is in MBASIC, there is a limit to what can be in memory and disk, thus the stories are indeed SHORT and very limited to response. When a question is presented the program looks for certain key words in your response, and prepares its answer based on this. There appears to be no random number generator to give you a different story every time, thus if you go through them again with the same answer, you get the same story exactly--this can get boring after a little while. I think the idea is clever, but was disappointed at the limitness for a \$17.50 pricetag, although it is to a point entertaining. I may have not been the best guy to review this, since I graduated from games and adventure type programs long ago - not because I don't enjoy them, but have no time for them.

Another story program from Evryware which I must rate higher is the interactive fiction's **DRAGONS OF HONG KONG** for \$21.50. This is only one story, presented in chapters. It is much longer than the above mentioned ones, since it is the only one on the disk. It is more interesting and interactive, with more possibilities. You can go many ways in this story, and can even get a second chance to 'rewrite history' if you flub up in certain parts. Unfortunately, I did not finish the story in 'the arms of the woman of my dreams', but since I'm married, that was probably for the best! This one works the same as the above by expecting certain 'key answers' in response to questions and finishing lines. I feel if you want to get into this type of thing, this one is better than the **SIX MICRO-STORIES**.

I want to introduce to you my software reviewer, Ralph Munroe, a computerist reviewing software with me for this newsletter. Ralph evaluates all software on an H8/H19 system operated at 19.2 K baud, and sometimes a Z89. Ralph's policy is to review software for publication in this newsletter and present points that the advertisements and mentions do not, like first hand user experience and impressions. These are good coming from others than myself all the time, and this helps free my time somewhat. Ralph operates his own printing and tape recording business and works for the Moody Bible Institute-- a busy guy. Most of Ralphs reviews will be printed practically verbatim, I will only edit to conserve space. Rating at end of each are combined ratings of all reviewers.

Ralph did a series for this issue on software from Evryware, so I'll print that right here.

Y-WING FIGHTER-- This game reminds me of the Defender arcade game, even though it is markedly different. One must guide his Y-Wing fighter over 560 miles of enemy planet's rugged terrain, shooting down (with your front and rear phasers) giant birds, enemy fighters, helicopters, etc., and avoiding anti Y-Wing fighter missiles, volcanoes, and other hazards and surprises.

A couple of teens that have "exercised" this software

have completed the 560 mile journey after invoking some real ingenuity, only to be shot down in sight of the enemy and his hide-away squadron of enemy fighters guarding his lair. None have succeeded in "terminating" the enemy and so the game continues to be a formidable challenge to one and all. . . **RATING A.**

GALACTIC WARRIOR-- Move over Invaders and make room for Galactic Warrior. For those who would like to see what they're up against, run this game in its demo mode and witness the first 4 enemy attack phases. You will see that you have 4 things going for you - lateral movement of your ship, a protective force field, a short range laser and deep space missiles. Your ship can take 2 direct missile, star bomb, or delta ray hits from the enemy interceptors, special ships or giant insects before being reduced to space dust by the 3rd hit.

The object of this game is to wipe out the enemy fleet so you may reach their space station, blast through its hull and score a direct hit on its central reactor. As with most of their software, the highest score is saved for recall by the next game. **RATING B.**

INVASION-- This game is patterned after the arcade game of the same name, but doesn't attempt to recreate the arcade game alien creatures as does the Software Toolworks version, but rather creates them with a personality all their own, complete with blinking eyes! Be careful though, to dodge not only the invaders' bombs, but also their radioactive wreckage that falls after you destroy them.

This software also features a demo mode by which you may familiarize yourself with the game. Although this game doesn't appear as challenging as the Software Toolworks version, just see how long you can last. This game does have a Pro and Beginner option. **RATING C.**

MISSILE CONTROL-- The player is defending 6 cities against enemy missiles, planes and their bombs with two missile launch pads on opposite sides of the screen. The position of the cross-hair for the launchers is controlled by the numeric keypad. The firing of them is controlled independently by the "A" and "S" keys on the keyboard (you need both hands). Eventually you'll lose the battle, but the longer you hold out, the higher your score. The game is more challenging the longer you play and at least at one point, you are given a reprieve and are returned everything you lost.

This is an excellent game that could have been enhanced by including provision for retaining the high score and scorer information. **RATING C.**

These and others are available from EVRYWARE/ POB 60802/ Sunnyvale, CA 94088.

REQUESTS

Y. Lee/ 1824 Save Ct/ Timonium, MD 21093 is interested in using a **plotter** with his H89. If there is anyone that can steer him in the right direction, he would appreciate hearing from you.

James Czebiniak/ 199 Vly Road Ext/ Schenectady, NY 12309/ (518) 869-8723 is trying to find a supplier of refill ribbons for the MX-80 cartridges in colors other than black.

David Nordquist/ 4708 Downey N.E./ Albuquerque, NM 87109 is looking for a program similar to the HDOS TEST17, but one that will work on the 80 track DS drives. He is looking for a general disk drive checkout program (See OFFERS). If anyone has such a program under HDOS or CP/M notify David, or better yet, myself for

publication.

Frank Adams/ 8 Autumn Ridge Rd/ Weston, CT 06883/ (203) 226-4996 is interested in finding a second source for the H19/H89 green screen CRT tube. Heath wants \$75, and he feels this is excessive.

Gary Hoffman/ 1502 Spillers Ln/ Houston, TX 77043 is interested in information on a SuperCalc newsletter or user group. Anyone know of this, please contact Gary.

David Orosz/ 1251 Canton Dr/ Milpitas, CA 95035, is trying to locate a hand-held optical scanner (similar to the ones Sears or Penny's use at their cash registers) that can read NON-OCR characters and input them into a file using PIE. This would help him in his business as a editor and proof reader. The way it is now, he must retype all text submitted to him by hand. If anyone has ideas or more information, please contact David.

Walter Scott/ 7608 Luscombe Dr/ Knoxville, TN 37919/ (615) 690-3864 is looking for a spelling program that will work with AutoScribe. I'm not sure if Software Toolworks **SPELL** will work or not, perhaps the format may have to be changed first. If anyone has such a spell checking program operational for AutoScribe files, please contact Walter.

WHO'S WHO

K. Hatcher/ 4735 Cowell Blv& Davis, CA 95616 is making a mention of a Heath/ Zenith division of a Software Lending Library. It is open to anyone wishing to take advantage of their services. There are no membership fees or dues of any kind. If you are interested in more information, send a SASE for a listing of available software. The object is to allow you to test new software and decide if you want to buy it, rent it, or forget it.

The **Pomona, CA HUG** meets the fourth Thursday of each month at the HEC in Pomona, CA at 7:30 PM. For more info, write POMONA HUG/ ATTN: H. Friedman/ 1555 N Orange Grove Ave/ Pomona, CA 91767.

William Velten/ 12685 Parktrails Ln/ Florissant, MO 63033/ (314) 355-4072 has available a short MBASIC program to POKE the output of the PC-12 hardware clock into memory and automatically set the correct time on the software clock CK.DVD. This can be accomplished on BOOT or made part of a prologue. Send SASE if you desire a copy. The PC-12 is the clock/parallel board for the H8 (this would probably work for the PC-89 as well, for those using the H89 version from Environmental Control Systems by Don Long.

William also has developed several programs applicable to a doctors office. He has an excellent appointment keeping program which includes Search and Find routines, programs for automatically completing the Illinois Public Assistance forms, etc. If anyone has a need for this, write to Bill. He says it is very difficult to find useful medical related software (anyone help?). For the price of 2 disks and postage, he will send them to anyone interested. They run under HDOS 2.0 and MBASIC 4.82 and require two or more drives. Please send your business card when writing.

He has also extensively rewritten the CALCU8.MB program which was one of the H-SCOOP offers. It is now much improved. Anyone wishing a copy send the original H-SCOOP disk along with a buck for postage.

Ralph Stoltz has a high recommendation for John Priebe/ 4804 Mt. Airy Rd/ Sylvania, OH 43560/ (419) 882-3626. John is the president of THUG, and Ralph considers him a

Software Expert. Ralph wanted to use his H19 keyboard as an ASK type keyboard (different configuration than the qwerty board). He said both Jim Evans (no address supplied) as hardware expert, and John as software expert solved his problem.

TECHNICAL FORUM

CP/M 2.2.03 SPELLBINDER PATCH

I received a patch for the early release of Spellbinder for use with Heath CP/M 2.2.03. Spellbinder worked OK with 2.2.02, and the new update Spellbinder works OK with 2.2.02 or 2.2.03 CP/M. This mod is for the IOS in the old Spellbinder to work with the NEW Heath 2.2.03 CP/M. That way you won't have to update Spellbinder if you don't want to, and still use Heath CP/M 2.2.03. I have not tried this patch so can't vouch for it, you'll have to see for yourself.

In the SPELLBINDER IOS table, locate label **TCMP**. It should presently be a 0, change it to 1, and re-assemble. Recompile (you can use Spellbinder to read and change the IOS if you desire, but be certain to use the /1/2 switches for the G and GD commands) the IOS. Full instructions are in the Spellbinder manual for doing these types of things. If you feel comfortable with CP/M Assembler, go ahead and try, else update your Spellbinder.

USING H8 Z80 BOARD

This info sent in by Milt Krauthoff/ N89 W16800 Cleveland Ave/ Menomonee Falls, WI 53051.

"The instructions received from Heath when you buy their Z80 CPU board advise you to modify the disk controller board, and to configure a memory board to start at zero instead of Heath's normal 8K starting location.

"After reviewing their instructions it becomes apparent that you would then have full use of all memory that you have when running CP/M and that you would lose 8K when running Heath HDOS.

"Heath's modification calls for removing capacitor C-15 and replacing it with a jumper wire, then also removing resistor R-6 from the controller board.

"Instead of making these mods, I mounted a DPDT switch on a small aluminum bracket on the rear side of the controller board (there is a convenient hole for this bracket already in the board, it is right near C-15). Use a small bracket, making sure that it does not short any PC traces.

"A little study will show that this switch can be wired so that in one position (for CP/M) it can short out capacitor C-15, and in that same position resistor R-6 can have one end lifted and brought to this switch so the resistor is out of the circuit. Then in the reverse position (HDOS) the switch can remove the short from C-15 and also connect R-6 back into the circuit, thus returning the circuitry back to its original configuration.

"I also installed a DIP switch on one of my Heath 8K memory boards so it could become the top 8K of memory (for HDOS) or the 0-org of memory for the CP/M. (this will of course only work for having at least 1 8K board (and bank select if a 64K board--ed)).

"Now, just by lifting the lid on my H8, I can reach in and switch the new slide switch on the controller board and the dip switch on the memory board and have the full 56K of memory in either HDOS or CP/M. It works great.

(there is plenty of room on the left side of the H8 to add external switches, but care must be taken of the signals taken there, and the interference levels--ed).

"If you have the older style of controller board (the one without the IC #U28) it appears that where you are told to "just lift the pins on U22" you could instead wire them to a similar DP switch to then put Heath's mod in and out of the circuit, thus accomplishing the same result."

USER FEEDBACK

Based on past articles in this column I have received some feedback which I felt appropriate to print.

From Ralph Munroe: **HUB REINFORCEMENTS:** "A year ago three production disks Ralph Munroe uses almost daily were beginning to show signs of twisting around the hub. Application of hub reinforcements has stopped this physical distortion and the same disks are still being used a year later with no loss of data.

I'd like to add to this by quoting from a Dysan disk technical manual. "To accommodate poorly designed drives, reinforcements are sometimes used on the diskette hole. The placement of these reinforcements can change the diskette's center hole position. In addition, the adhesive used to affix the reinforcement must not bleed and gum up the heads. For these reasons, if they are required, it is best the media manufacture affix the reinforcements."

I also want to add my own caution. Yes, be very careful of the hub reinforcements. I purchase mine from Pacific Exchanges and have been very happy with them. I once purchased a batch (never again!) from the local Heath store, and threw them away. They were sticky on both sides, and indeed when affixed did cause the adhesive to bleed out. It really messed up the diskette. I made a bad mistake of giving this to a local customer I support. I had to make a free service call (there's \$30 down the drain) to open a disk drive door which had become stuck to the diskette. When the secretary tried to open the door, the disk stuck and got all bent up inside the drive, preventing the door from opening. Quite a job! Whoever made those reinforcements, had the right idea, but did not know what they were doing. Don't buy these! If they appear sticky all over and are very hard to separate, return them for your money and report the seller to your local computers against assassin's club. If that does no good and you have a bomb, write me and I'll send you some matches.

On **DRIVE LUBRICATION** Ralph writes "Shortly after reading the articles in H-SCOOP about lubricating the screw-type stepper head system (2/82 issue, p 6-7; 4/82 p 2,8), I had a drive start howling." Ralph called the local gun shop and found they carried TRI-FLON a spray teflon lubricant, and BREAK-FREE a liquid lubricant in a bottle, both teflon based. Ralph used the BREAK-FREE, since he felt since it was not a spray product he'd have better control over it. (I use the spray, but first apply the spray to a cotton swab or screwdriver tip--ed). "After applying BREAK-FREE to the stepper screws and head guides and 2 drops of 3-in-1 oil to each of the stepper motor bearings as recommended in the April '81 H-SCOOP, page 9, I was astonished as to how quiet the H17 drives could be."

Feedback from Donald Simpson: "...Accordingly here are two of my recent problems and fixes. (You suggested the fix to this one over the phone). **Problem:** Several characters permanently changed. **Fix:** Look at schematic. If characters are all on the same line to the Character Generator Chip, then replace the chip." (This was in response to a question called in by Don. He was hitting certain keys and they would display something other than what they should have on the screen. I

checked the schematic and suggested to him to check his keyboard encoder chip--ed).

"Problem: Disk read failures, usually on the same file. At first I thought this was being caused by bad disks. The same error always occurred on the same disk file. When I got a hard read error on the same file on two backups and the original distribution disk it occurred to me that this was stretching the laws of probability a bit. Using Heath's suggestion I took everything apart and looked at it. This brought results. When I got it all back the system wouldn't even boot. Once more everything out on the table. Now I could see something wrong. When the SY0: drive slides into the cabinet the little five prong plug on top can get pulled from its socket. This had happened but I also noticed that a transistor on the front middle of the circuit board had been broken. Replacing this and of course plugging the plug back completely cured the problem."

Let me say this about that. I too have had a drive with a transistor broken off the board. Plugs coming off are also a big problem. Both of these are especially prone to happen with much drive swapping or anytime drives are taken in and out of the cabinets a lot.--ed

And lastly, the very last word on flipping disks in a drive to use both sides. You all know my stand on that, and some wrote in saying that's not true, and I was too harsh. Again, from the Dysan diskette technical manual. **"Flipping a Diskette"** To utilize both sides of a diskette on a one sided drive, some users and manufacturers have modified the diskette so it can be "flipped." This technique has several drawbacks, all of which can result in a permanent data loss.

"First, in normal operation the drive pressure pad and diskette lining gather contaminants. When the diskette is flipped its rotation reverses and these contaminants are jarred loose, resulting in coating abrasion and head contamination. In addition, the contaminants collected on the pressure pad abrade side 0 when the diskette is flipped.

"Second, the coating takes a set corresponding to the direction of rotation. This is similar to the set that occurs with automobile radial tires. Moving a radial tire from the right to left side of a vehicle reverses its direction of rotation. Rotation in reverse can result in structural damage, radically decreasing the tire's lifetime. The same effect can be observed on a flipped diskette.

"Third, a diskette tends to develop a grain corresponding to the direction of rotation. Flipping a diskette results in the head going against the grain. This can result in premature head wear and loss of head magnetic properties due to friction overheating.

"Considering the relatively high cost of drive repairs, the low cost of diskettes, and the incalculable worth of stored data, it is economically unwise to advocate flipping diskettes. . ." Let's bury the hatchet (and handle) on this topic.--ed

CLASSIFIEDS

FOR SALE--Anderson/ Jacobson AJ-841 printer. Like Selectric II, except RS-232 Serial at 15 cps. \$800 with tractor feed & manuals. Like new. Ray Albrektson/ 619 W Dover/ San Bernardino, CA 92407.

FOR SALE--Three 8K Heath static memory boards for the H8, \$45 each or all 3 for \$125. H8-5 serial/cassette board, \$60. CAT Modem, \$75. H-9 Terminal, no mods, \$110. I will pay shipping to continental US. Frank Adams/ 8 Autumn Ridge Rd/ Weston, CT 06883/ (203) 226-4996.

FOR SALE--Heath 8080 CPU card for H8 computer, 0-org card for H8, and tin motherboard for H8. All with documentation. \$100 for all. Jim Benesch/ Box 1811/ Minden, NV 89423/ (702) 266-3686.

FOR SALE--4 Heath 8K RAM boards for H8 at \$40 each. 2 H8 16K RAM boards for \$95 each. Teletype Model 43 for \$550. Mark Wollin/ 115 W Oak St/ Lake Mills, WI 53551/ (414) 648-8283.

FOR SALE--Bare H8 computer for \$150, will consider trade for used disk drive. Trionyx 64K memory board fully populated with memory chips for \$250. H8-5 Serial and cassette controller for \$25. Will sell the entire above package for \$350. Also have H88-5 Cassette interface board for the H88 or H89 for \$20. Thomas H Bessire/ 7600 Pacific Place_ Ft. Worth, TX 76112/ (817) 451-3912.

FOR SALE--Junkbox parts bonanza. 12 - 10 X 14 boards from a mini computer loaded with IC's, tantalum capacitors, etc. Each board contains between 15 and 100 IC's. Removed from mini in hospital environment. Gold edge connectors broke off board. \$4 each, 3 for \$10. 19 - same as above but with gold edge connectors for \$5 each or 6 for \$25. 1 rack mount card cage complete with mother board for above cards, holds 19 cards (16 X 11 X 19) for \$25. I pay shipping on all orders over \$25, smaller orders add \$3 shipping per order. Contact H-SCOOP.

FOR SALE--H8 with 0-org; H8-4 4 port serial board; H8-5; Trionyx 64K RAM card; H17 disk controller board; 2 H-17 40 track drives; H19 fabric screen terminal; IDS 440 Paper Tiger dot matrix printer (2K graphics buffer). Price is \$3000 complete. Richard Melvin/ 2103 First St/ La Grande, OR 97850/ (503) 963-8950.

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PHONE HOURS: 9AM to 9PM as I am available, except sunset Friday to sunset Saturday (Seventh Day Adventist). NO COLLECT CALLS.